NAME : SANDESH NAKADE

CLASS : CSAI-D

ROLL NO : 34

PRN :M 12412601

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_score, f1_score
```

```
df = pd.read_csv("/content/cosmetics_shop_dataset_with_discount.csv")
print(df.head())
```

```
   Age  Gender  Monthly_Income  Visit_Frequency_per_Month  \
0   56    Male           51631                          5
1   46  Female           87991                          2
2   32  Female           19014                          8
3   25    Male           26093                          4
4   38  Female           33070                          9

   Avg_Purchase_Amount Membership Previous_Offer_Response  \
0                 3478        Yes                     Yes
1                 1452         No                      No
2                 4181         No                     Yes
3                 4873         No                     Yes
4                 2322        Yes                      No

  Response_to_Special_Offer  Discount_Percentage
0                       Yes                   25
1                        No                    5
2                       Yes                   19
3                        No                   16
4                       Yes                   19
```

```
df.shape;
```

```
le = LabelEncoder()

for col in df.columns:
    if df[col].dtype == 'object':
        df[col] = le.fit_transform(df[col])
```

```
X = df.drop("Response_to_Special_Offer", axis=1)
y = df["Response_to_Special_Offer"]
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
  n_iter_i = _check_optimize_result(
```

```
▾ LogisticRegression
LogisticRegression(max_iter=1000)
```

```
y_pred = model.predict(X_test)
```

```
cm = confusion_matrix(y_test, y_pred)
print(cm)
```

```
[[15  2]
 [ 2 21]]
```

```
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

print("Accuracy:", accuracy)
print("Precision:", precision)
print("Recall:", recall)
print("F1 Score:", f1)
```

```
Accuracy: 0.9
Precision: 0.9130434782608695
Recall: 0.9130434782608695
F1 Score: 0.9130434782608695
```